

Technology Trends and Insights Report

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OUTLINE



- Executive Summary
- Introduction
- Methodology
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 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- This presentation will display trends in both programming languages and databases.
- Demographics survey.
- Future trends in technology, consisting of languages, databases, web frames and platforms.
- Technological gaps in differing countries.

INTRODUCTION



- **About: Analyzing trends in IT skills and development tools**
- **Purpose:**
 - Presentation will provide insights into future skill requirements
 - What software development skills are most in demand?
 - What programming languages are most in demand?
 - What are the differences in demographics?
- **Audience: IT head**



METHODOLOGY



- **Resources for data collection:**
 - Stack overflow developer survey
 - GitHub job postings
 - Developer tools annual compensation
- **Data exploration**
- **Data wrangling**
- **Data visualization**
- **Presentation of key findings**



RESULTS

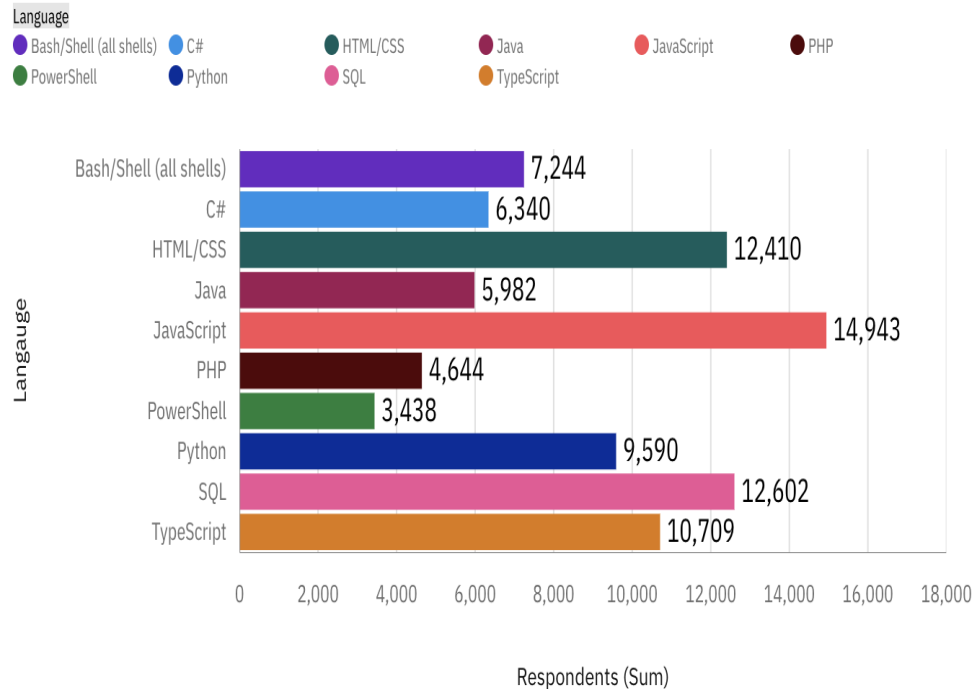
ResponseId	LanguageHaveWorkedWith	LanguageWantToWorkWith	DatabaseHaveWorkedWith	DatabaseWantToWorkWith
0	2 Bash/Shell (all shells), Go, HTML/CSS, Java, J...	Bash/Shell (all shells), Go, HTML/CSS, Java, J...	Dynamodb, MongoDB, PostgreSQL	PostgreSQL
1	3 C#	C#	Firebase Realtime Database	Firebase Realtime Database
2	10 HTML/CSS, JavaScript, TypeScript	HTML/CSS, JavaScript, TypeScript	MongoDB, MySQL, PostgreSQL, Redis	MongoDB, MySQL, PostgreSQL, Redis
3	11 JavaScript, Python, Ruby	C#, Crystal, Java, JavaScript, Python, Ruby	Elasticsearch, MongoDB, PostgreSQL, SQLite	Elasticsearch, MongoDB, PostgreSQL, SQLite
4	12 C, C#, C++, Delphi, HTML/CSS, Java, JavaScript...	C#, HTML/CSS, JavaScript, PowerShell, Rust, SQ...	Microsoft Access, Microsoft SQL Server, Postgr...	Microsoft SQL Server, PostgreSQL, SQLite
5	19 C#, HTML/CSS, JavaScript, PHP, Python, SQL	C#, HTML/CSS, JavaScript, SQL	Microsoft Access, Microsoft SQL Server, MySQL,...	Microsoft SQL Server, SQLite
6	23 Bash/Shell (all shells), C#, F#, JavaScript, P...	Bash/Shell (all shells), C#, F#, Go	Redis, SQLite	Redis, SQLite
7	24 HTML/CSS, JavaScript, PHP, Python, SQL	HTML/CSS, JavaScript, PHP, Python, Rust, SQL	MySQL, PostgreSQL, SQLite	MySQL, PostgreSQL, SQLite
8	27 C#, HTML/CSS, JavaScript, Python, Scala, SQL	C#, JavaScript, Python, Scala, SQL	Microsoft SQL Server, PostgreSQL, Presto	PostgreSQL, Presto
9	29 Bash/Shell (all shells), C#, HTML/CSS, JavaScr...	Bash/Shell (all shells), C#, HTML/CSS, JavaScr...	MariaDB, Microsoft SQL Server, MySQL	MariaDB, Microsoft SQL Server, MySQL

Results based on this table

PROGRAMMING LANGUAGE TRENDS

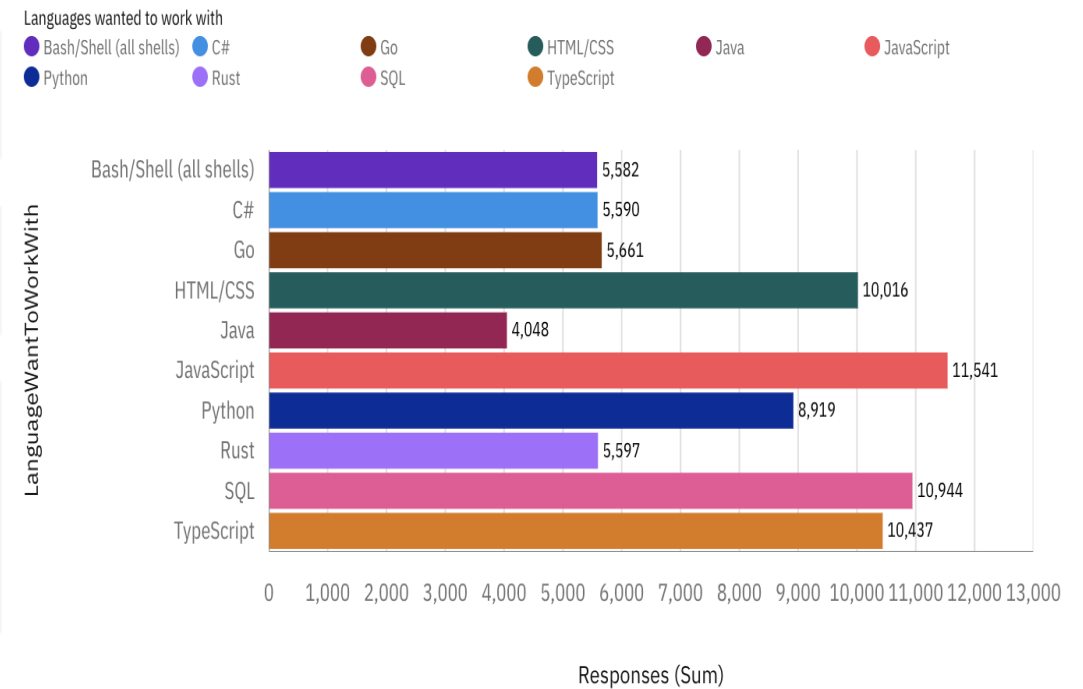
Current Year

Top 10 languages worked with



Next Year

Top 10 Languages developers want to work with



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, HTML/CSS and SQL are the top 3 for this year.
- The 3 above are becoming less popular next year.
- TypeScript and Rust becoming increasingly popular.

Implications

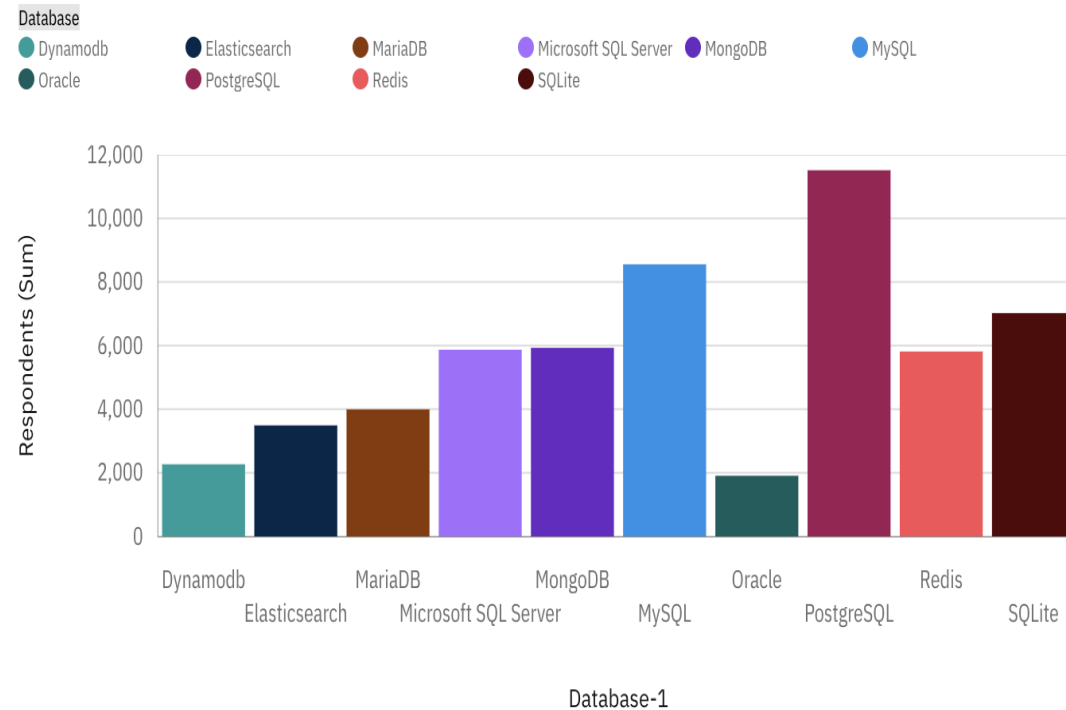
- Web development applications still in high demand.
- Training should focus on both core and emerging languages.
- Employers may need to adapt tech stacks gradually.



DATABASE TRENDS

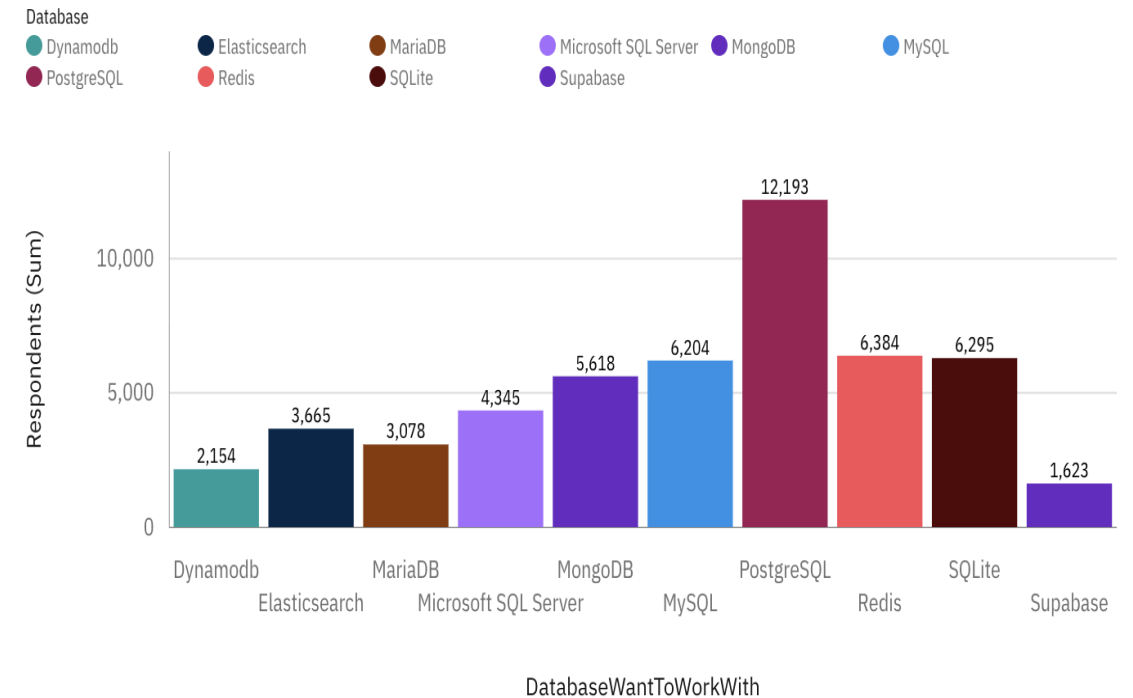
Current Year

Top 10 databases worked with



Next Year

Top 10 Databases developers want to work with



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL is the most popular database**, leading both current use and future preference.
- **MySQL and SQLite are widely used currently**, but show lower interest for next year compared with PostgreSQL.
- **Supabase appears in future preferences**, suggesting growing interest in newer cloud/backend database platforms.

Implications

- **PostgreSQL skills should be prioritized**, as it is both widely used and highly desired by developers.
- **Organisations should maintain support for MySQL and SQLite**, but expect some movement towards PostgreSQL and other modern tools.
- **Training should include emerging database platforms**, especially cloud-based tools such as Supabase alongside traditional SQL databases.



DASHBOARD

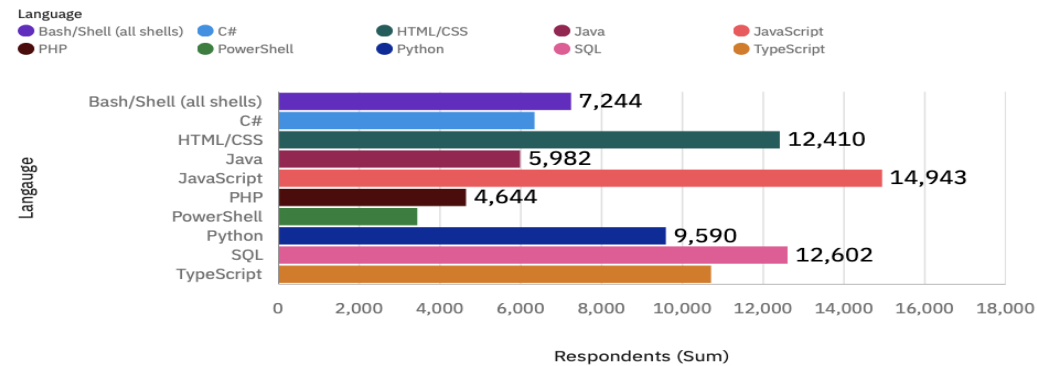
[IBM Cognos Analytics Dashboard link](#)



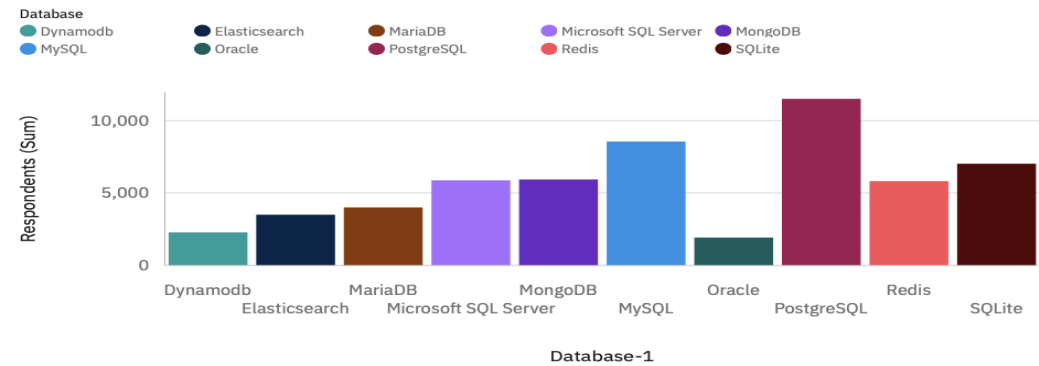
DASHBOARD TAB 1

Current Technology Usage

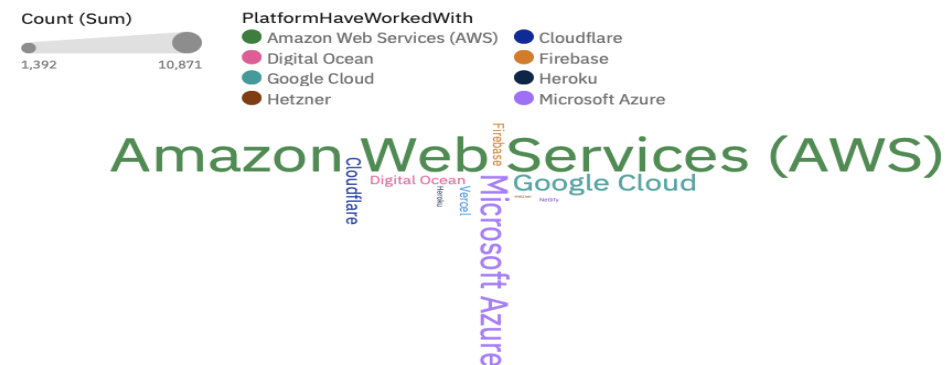
Top 10 languages worked with



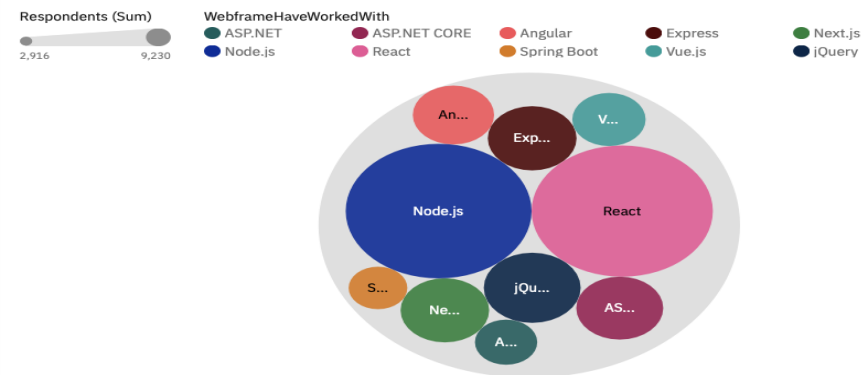
Top 10 databases worked with



Top 10 platforms worked with



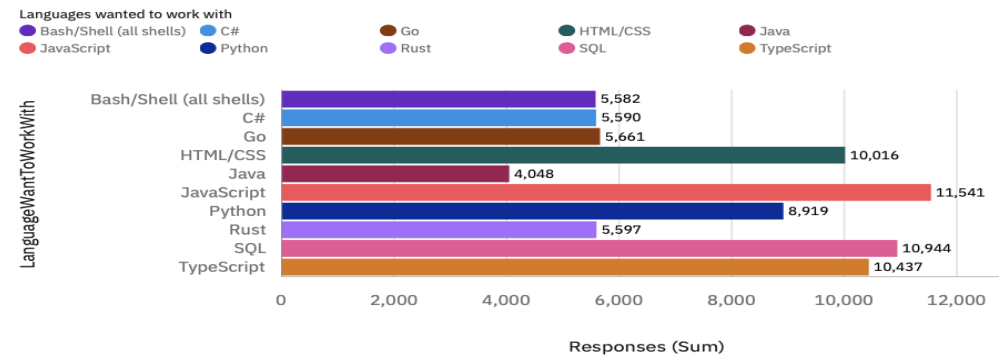
Top 10 web frames worked with



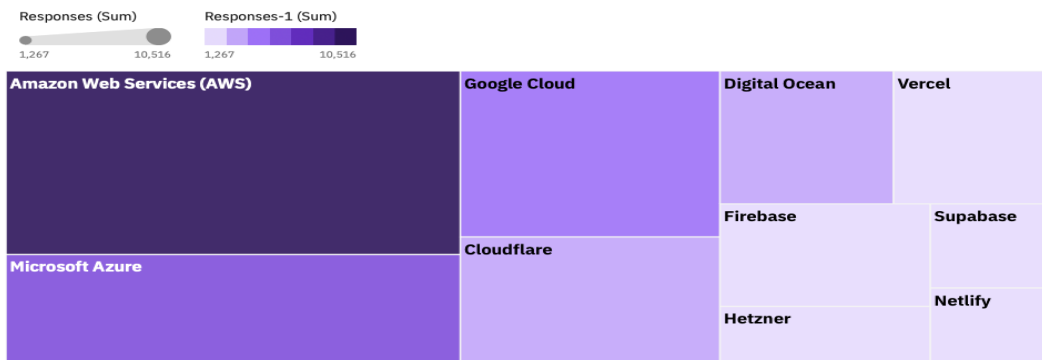
DASHBOARD TAB 2

Future Technology Trend

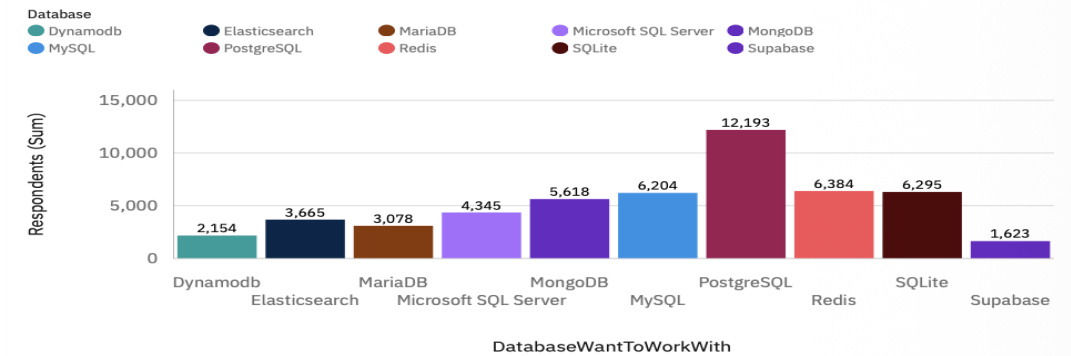
Top 10 Languages developers want to work with



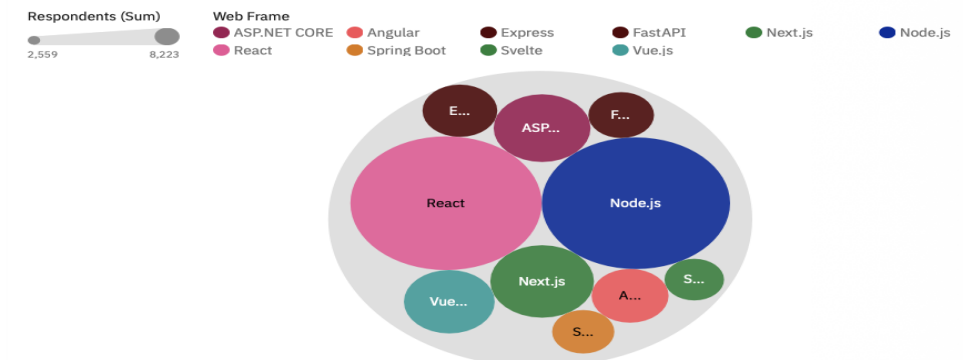
Top 10 Platforms developers want to work with



Top 10 Databases developers want to work with



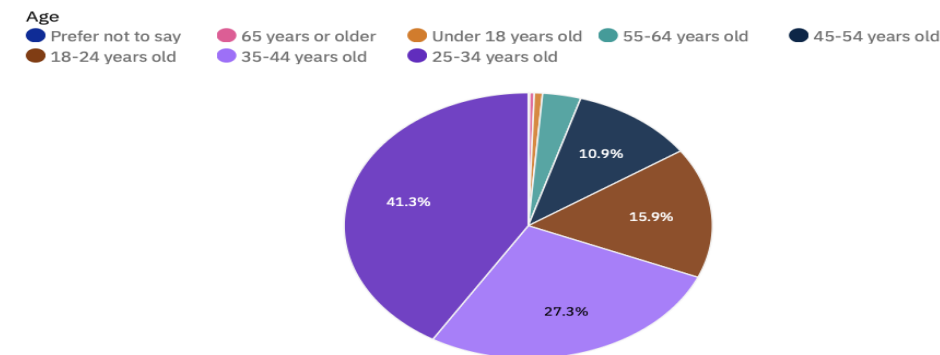
Top 10 Web frames developers want to work with



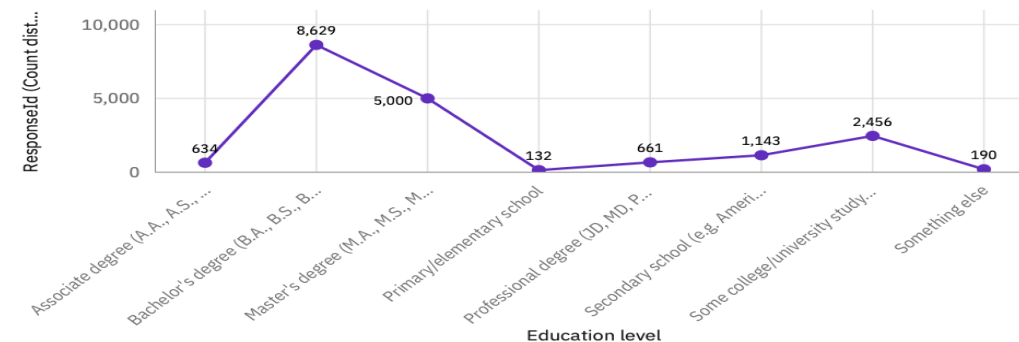
DASHBOARD TAB 3

Demographics

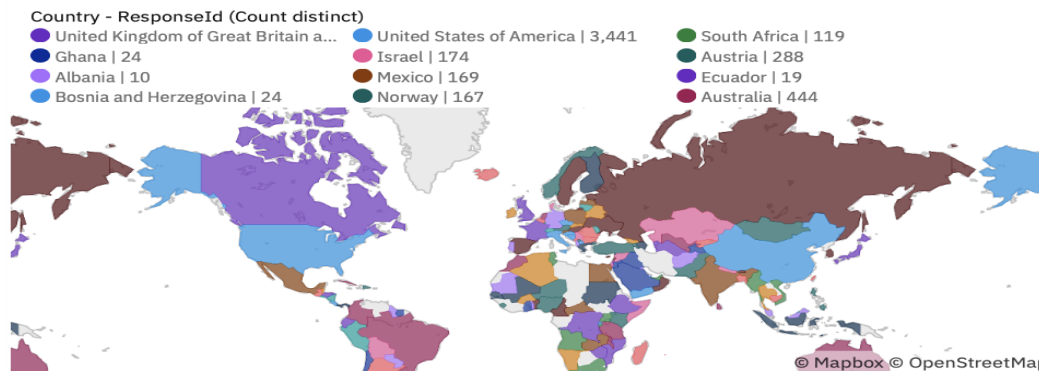
Respondent distribution by age



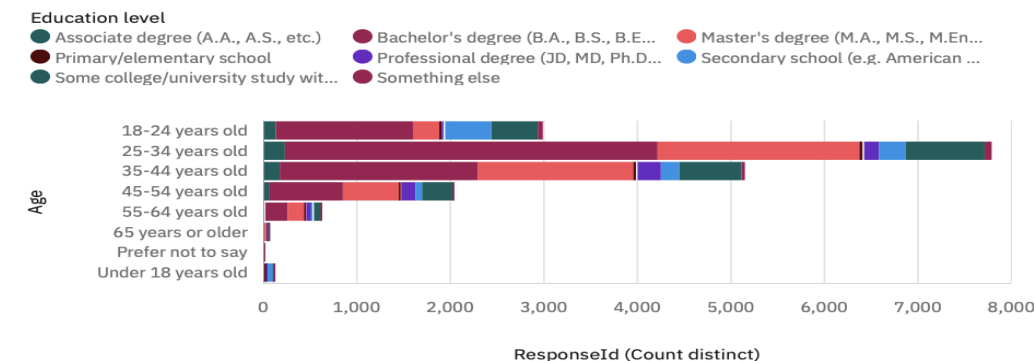
Respondent distribution by Formal Education Level



Respondent Count by Country



Respondent Count by Age, classified by Education Level



DISCUSSION



- Trends for now and in the future
- Training that should be provided to tech workers
- Upskill tech skills for those in developing nations
- Reduce age discrimination in employment

OVERALL FINDINGS & IMPLICATIONS

Findings

- Technologies used are changing at a rapid rate
- Concentration in developed nations, such as the US and the UK
- Programming languages such as TypeScript and Rust are becoming increasingly popular

Implications

- Companies need to adjust to rapid changes in the tech landscape
- Impact awareness of AI automation
- Become more aware of cloud services.

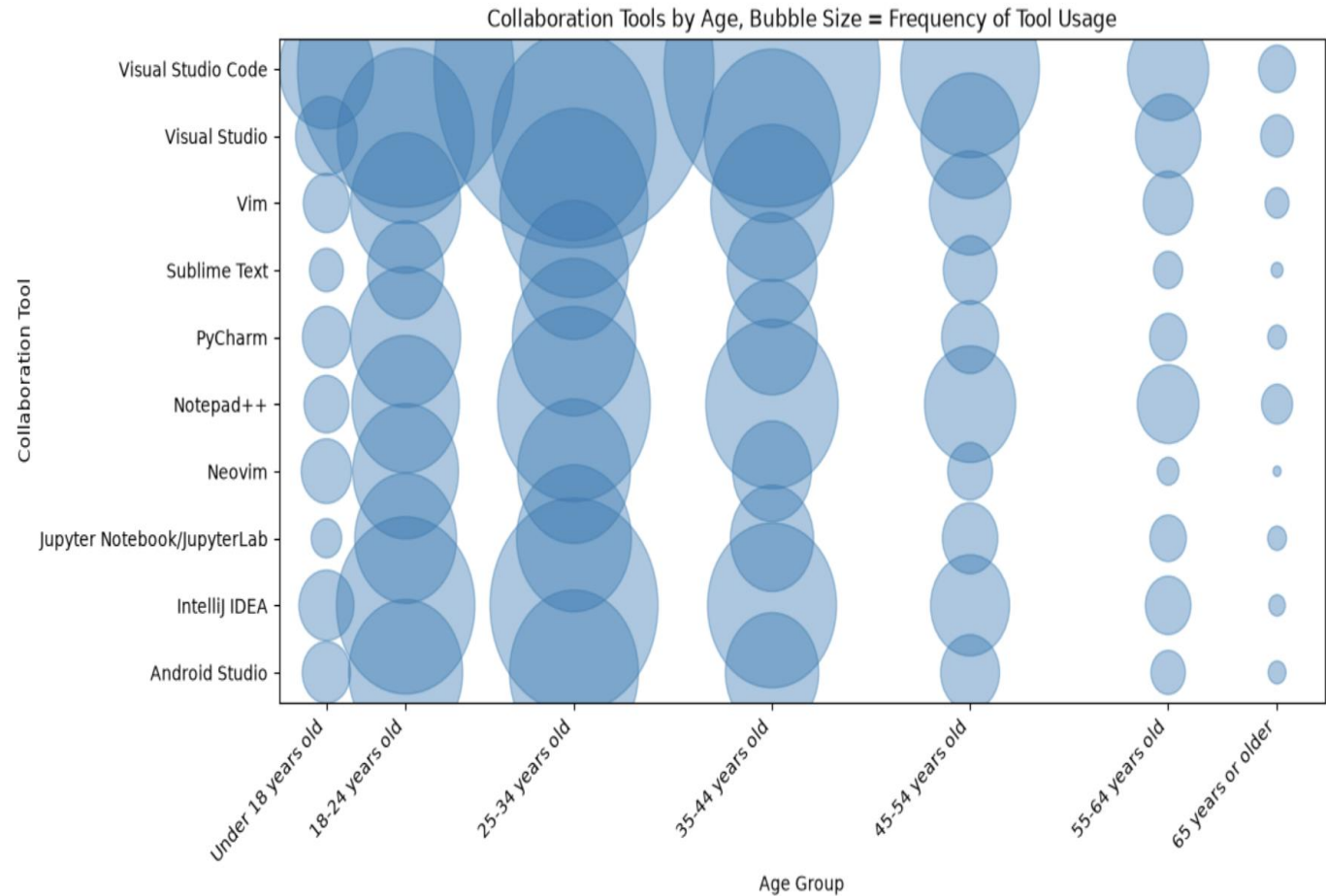


CONCLUSION

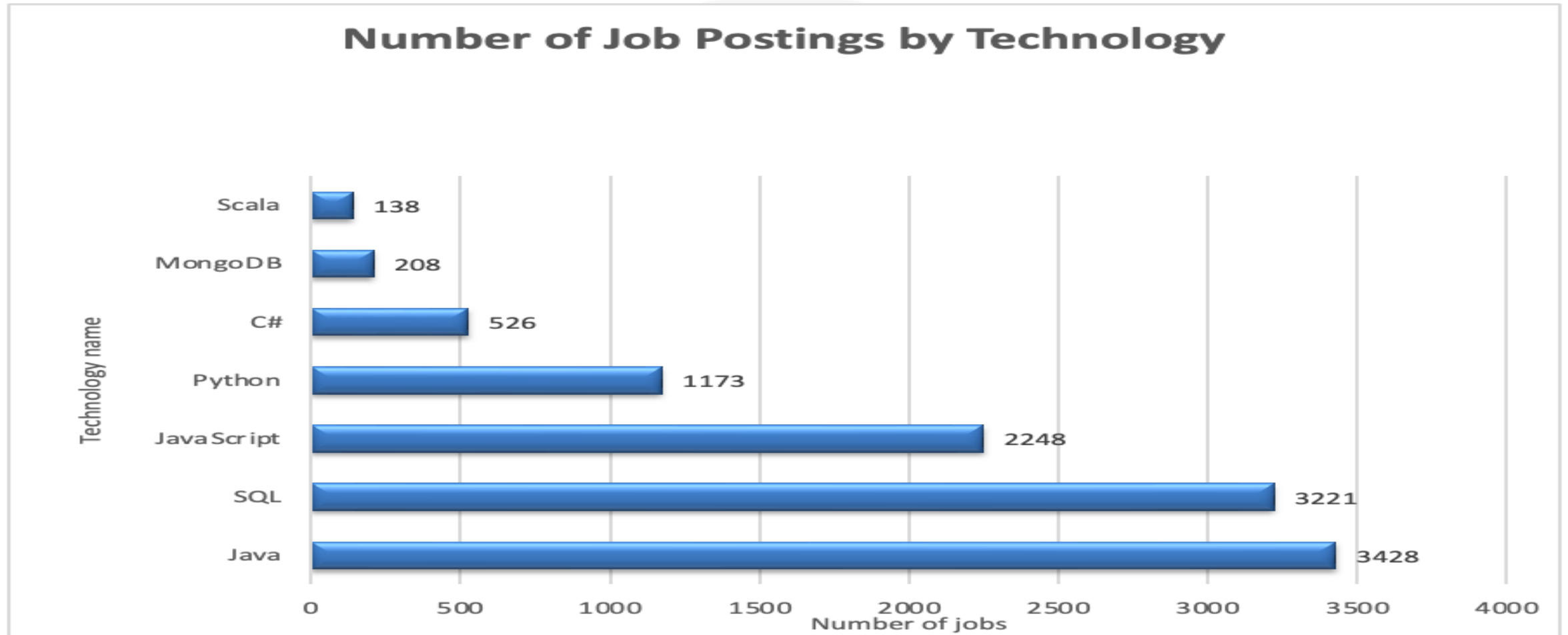


- **Core web development skills remain essential**, with JavaScript, HTML/CSS, SQL and Python continuing to dominate current developer usage.
- **Future demand is shifting towards emerging and modern tools**, including TypeScript, Rust, PostgreSQL, Supabase and cloud-based platforms.
- **Technology training should balance established skills with future-focused upskilling**, helping developers stay relevant as tools and frameworks change.
- **Demographic and geographic differences highlight the need for inclusive access to tech education**, particularly across countries, age groups and education levels.

APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

